#BernoulliRandomVariable, #BinomialRandomVariable, #Probability, #Randomvariable, #Statistics

Notebook

Day-5

Random Variables.

Random Variables.

Random Variables, X

$P(X)$ is a function called distribution
(a) Values of X
(b) probability p of each value.

Ex: Flip a coin 4 times.

$\Omega = \{hhhh, hhht, hhth, \ldots, tttt\}$

X = # heads that occur.

$(X \in \{0,1,2,3,4\})$ [discrete random variable].

Discrete = # of heads in n coin flips : $X \sim$ binomial

Continuos = height of americans. : $X \sim$ normal.

$P(X=0) = \frac{1}{16}$

$P(X=1) = \frac{4}{16}$

$P(X=2) = \frac{6}{16}$

$P(X=3) = \frac{4}{16}$

$P(X=4) = \frac{1}{16}$

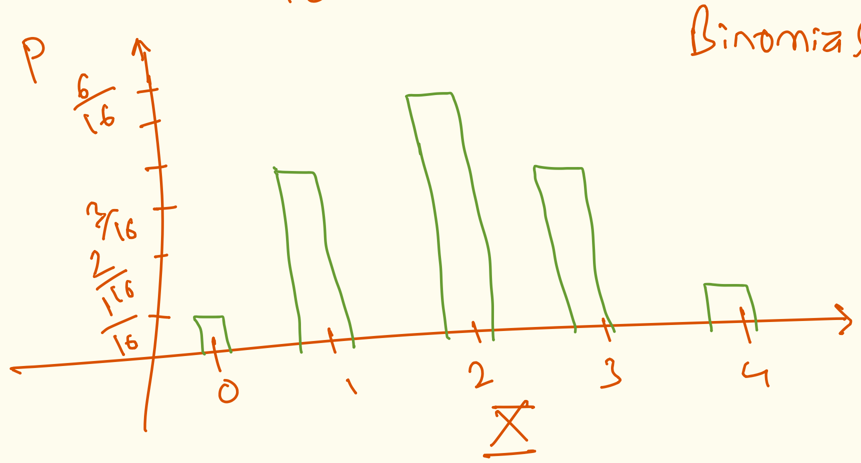
1, 4, 6, 4, 1

↓

Binomial Coefficients.

↓

Binomial distribution.



Why do we want random variables?

1) Concise summary of all probabilities.
(function for $P(X)$ ... no counting)

2) prob. Density Function (PDF) is a model of a random process.
(test hypotheses with data)
(parameter estimation with data)

3) Visualize and compute.
.... plot ... compute $P(4 \le X \le 6)$

$$\sum_{k=4}^{6} P(X=k) \quad \text{or} \quad \int_4^6 P(X)\,dx$$

4) Build functions on X
X^2 ... propagate uncertainty through dynamics.

Bernoulli & Binomial (random variables X)

A Bernoulli Random variable X can take two values

Range $= \{0, 1\}$

0 → did not happen

1 → does happen.

(Coin flip, getting Head, dice roll... get 6, part... 6)

$P(X=0) = 1-p = q$ (failure)

$P(X=1) = p$ (success).

Ex: For masking images randomly → can use a Bernoulli dist.

(dropout regularization in a neural network masks each unit with an independent Bernoulli draw)

If I repeated and added $\Rightarrow$ Binomial.

If n independent Bernoulli random trials are performed $(B_1, B_2, \ldots B_n)$

Let X = total # successes $= B_1 + B_2 + \ldots + B_n$

$$X \sim \text{binomial}(n, p) \Rightarrow P(X=k) = \binom{n}{k} p^k q^{n-k}$$

success (p^k), non-success (q^{n-k}).

Also, $\boxed{\sum_{k=0}^{K} \binom{n}{k} p^k q^{n-k} = 1}$

Large n

$n \to$ large

If np or nq small: $X \to \text{Poision}(\lambda)$

If npq not too small, $X \to \text{binomial}(n,p) \to X \sim \text{normal}(np, \sqrt{npq})$

expected val. (mean) (np), S.D. ($\sqrt{npq}$)

Example: $n=2$

number of success — no success at all — failure on 1st trial

$$P(\underline{X}=0)=P(F_1F_2)=P(F_1)\cdot P(F_2)=q^2$$

1 success — success on 1st trial

$$P(\underline{X}=1)=P(S_1F_2)+P(F_1S_2)$$
$$=P(S_1)P(F_2)+P(F_1)P(S_2)$$
$$=pq+qp=2pq.$$

2 success.

$$P(\underline{X}=2)=P(S_1S_2)=P(S_1)P(S_2)=p^2.$$

Example:

DIY: $n=3.$

$$P(\underline{X}=0)=P(F_1F_2F_3)=P(F_1)P(F_2)P(F_3)$$
$$=q^3$$

$$P(\underline{X}=1)=P(S_1F_2F_3)+P(F_1S_2F_3)+P(F_1F_2S_3)$$
$$=P(S_1)P(F_2)P(F_3)+P(F_1)P(S_2)P(F_3)$$
$$+P(F_1)P(F_2)P(S_3)$$

$$=pq^2+pq^2+pq^2=3pq^2$$

$$P(\underline{X}=2)=3p^2q$$
$$P(\underline{X}=3)=p^3$$

Try adding: $p^3 + 3p^2q + 3pq^2 + q^3$

$$(p+q)^3 = 1^3 = 1$$

Ex: prob. of getting 3 sixes on 12 dice rolls.

$p = \frac{1}{6}$

$q = \frac{5}{6}$

$n = 12$

$X = 3$

$X \sim \text{binomial}(12, \frac{1}{6})$

$P(X=3) = \binom{12}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^9$

$= \_\_\_\_\_\_.$

$$P(3) = \binom{12}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^9 = \_\_\_\_\_\_$$

Small coded example regarding the use of Bernoulli distribution in Dropout regularization in an NN.

⇒ np.random.uniform(0,1) >self.p [threshold comparison].

1 → success → keep the neuron active.

0 → failure → drop it.

with (1-p) probability of survival.

Also, a scaling trick called "inverted dropout" is used sometimes which boosts the surviving activations at training time so their expected total magnitude stays the same whether or not dropout is active, meaning nothing needs to change at inference time.

(we will understand it once we get to Neural Networks).